

Domanda 1

Testo della domanda

VPNs (virtual private networks) are used to

Risposta

- a. Partition a private network (for example the one of a parent company with various subsidiaries) in multiple networks virtually separated
- b. Divide a corporate local area network in a set of separate subnets, each for a different corporate function (e.g., sales, procurement, engineering, marketing)
- c. Transport private traffic through a shared infrastructure while creating the same conditions the traffic would undergo through a private infrastructure

Domanda 2

Testo della domanda

The Layer 3 VPN (virtual private network) solutions based on MPLS are characterized by

Risposta

- a. A good level of automation and integration between the public backbone and private networks
- b. Particularly high security thanks to the deployment of cryptographic techniques
- c. Layer 3 tunneling mechanisms, namely within IP packets
- d. A mix of Layer 2 and Layer 3 tunnelling mechanisms

Domanda 3

Testo della domanda

The IPsec “Tunnel Mode” encompasses the encryption of

Risposta

- a. The IP header, TCP/UDP header, and payload of the internal packet
- b. Only the TCP/UDP header and payload
- c. Only the payload of the internal packet
- d. The whole external packet, header included

Domanda 4

Testo della domanda

Two hosts A and B belong to the same physical network and have IP addresses 130.192.1.1/25 and 130.192.1.129/24, respectively.

Risposta

- a. A directly communicates with B and vice versa
- b. A directly communicates with B but not vice versa
- c. A can communicate with B only by means of a router
- d. A cannot communicate with B

Domanda 5**Testo della domanda**

Which of these techniques is not a solution for the IPv4-IPv6 transition?

Risposta

- a. 6to4
- b. 6over4
- c. Teredo
- d. 6mix4

Domanda 6**Testo della domanda**

In the DS-Lite solution for the IPv4-IPv6 transition

Risposta

- a. The entries of the filtering database of an Ethernet switch
- b. The NAT feature is not available
- c. The NAT feature is implemented for all users on proper ISP devices
- d. The NAT feature is implemented on both the CPE and the ISP devices

Domanda 7**Testo della domanda**

The metric (cost) used by a routing algorithm

Risposta

- a. Expresses the probability to use the shortest path
- b. Expresses the complexity of the algorithm in performing the path computation

c. Expresses the weight assigned to a link (channel) in the path selection

Domanda 8

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BGP is used in the Internet for

Risposta

- a. Find out the geographic position of a host based on its IP address
- b. Discovering neighboring (bordering) routers on a local are network
- c. Communicating to neighboring routers the state of the links of a router
- d. The exchange of routing information between routers belonging to different autonomous systems

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Testo della domanda

The difference between link state and distance vector routing algorithms can be summarized as follows:

Risposta

- a. Link state algorithms send local information only to neighboring nodes; distance vector algorithms send global information to all nodes in the network
- b. Link state algorithms send local information to all nodes in the network; distance vector algorithms send global information only to neighboring nodes
- c. Link state algorithms send global information to all nodes in the network; distance vector algorithms send local information only to neighboring nodes

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Testo della domanda

RIP is characterized by

Risposta

- a. The suitability to both interdomain and intradomain routing
- b. The possibility to operate on large networks thank to its capability to function in a hierarchical way
- c. The usage of a link state routing algorithm
- d. Frequent instability and inclination to create circular forwarding paths (i.e., routing loops)

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Testo della domanda

Two IP networks 130.192.0.0/24 and 130.192.2.0/24 can be aggregated in:

Risposta

- a. 130.192.0.0/23
- b. They cannot be aggregated
- c. 130.192.0.0/22
- d. 130.192.2.0/23

Domanda 12**Testo della domanda**

The Integrated IS-IS protocol

Risposta

- a. Is a protocol used by Ethernet switches to create a routing tree in the network (spanning tree) removing closed paths (i.e., loops)
- b. Is a protocol derived as an evolution of BGP for the exchange of information between routers belonging to different autonomous systems
- c. Is a protocol based on the link state routing algorithm widely used in large networks
- d. Is an obsolete routing protocol no longer used due to its low performance

Domanda 13**Testo della domanda**

RSVP (Resource reSerVation Protocol) allows:

Risposta

- a. To monitor delay and loss experienced in the network by packets of multimedia applications
- b. Servers to reserve computing resources in their shared processors
- c. Routers to know the requirements of an application in terms of quality of service
- d. To limit the delay variation (jitter) experienced by packets in routers

Domanda 14**Testo della domanda**

In DiffServ, a “class of service” identifies:

Risposta

- a. A set of packets that are handled the same way by routers (for example, all VoIP traffic)
- b. A working mode of border routers that have to classify and mark incoming packets
- c. A set of packets that belong to the same VoIP session

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Testo della domanda

The importance of MPLS (multi-protocol label switching) in today's and future computer networks stems from the possibility to

Risposta

- a. Have a single control plan for different switching technologies
- b. Balance traffic across a server farm
- c. Realize devices that can operate without needing to be configured
- d. Realize switches with specific support to guarantee quality of service

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Testo della domanda

One of the protocols used in MPLS for label distribution is:

Risposta

- a. OSPF
- b. BGP
- c. L2TP
- d. IS-IS

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In the MPLS (multi-protocol label switching) architecture, LSPs (label switched paths)

Risposta

- a. Consist in the shortest path towards a given destination
- b. Represent alternative paths for forwarding packets towards a given destination kept by a router in its routing table
- c. Are set up by network nodes that agree on the labels to be used for packets belonging to a specific forwarding equivalence class (FEC)

d. Are exchanged by routers to create a map of the network

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The operations that an MPLS router can perform on labels are:

Risposta

- a. Labels cannot be modified by routers.
- b. Add a label in most external position of the MPLS header (PUSH), remove a label from most external position in the MPLS header (POP), change the content of the external label (SWAP).
- c. Add a label only if there are no others in the MPLS header (only one label is allowed) (PUSH), remove the only allowed label from the MPLS header (POP) upon the packet exiting the MPLS network, change the content of the label (SWAP).
- d. Add a label in any position of the MPLS header (PUSH), remove a label from any position in the MPLS header (POP), change the content of any label (SWAP).

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The IGMP protocol

Risposta

- a. Is a new version of the ICMP protocol
- b. Allows an IPv4 router to discover which multicast groups are present in a directly connected network
- c. Allows an IPv4 router to discover which multicast groups are active in the Internet network at a given instant of time
- d. Allows an IPv4 host to create a new multicast group

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In an IPv4 network

Risposta

- a. A host always delivers to the application layer all the multicast packets received
- b. A host is reached by a multicast packet related to a specific group only if it joined that group, whichever is the layer 2 technology adopted in the network
- c. A host can be reached by a multicast packet related to a specific group even if it did not join that group before

d. A host cannot understand a multicast IPv4 packet

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The Neighbor Discovery procedure in IPv6

Risposta

- a. Is based on a multicast ICMPv6 packet
- b. Is based on ARPv6
- c. Needs the network to support IPv4
- d. Is based on a broadcast ICMPv6 packet

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The ICMPv6 Router Advertisement packet

Risposta

- a. Enables device autoconfiguration without a DHCP protocol intervention
- b. Is sent periodically by a router to all the other routers of the Internet network
- c. Is a broadcast packet
- d. Is sent as a reply to an ICMPv6 Neighbor Solicitation packet

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IPv6 Private addresses

Risposta

- a. Are used only on routers
- b. Are defined in such a way that they are globally unique with high probability, but they cannot anyway be used on a global level
- c. Are used to interconnect private networks through a public network
- d. Are used only for on-link communications

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Testo della domanda

IPv6 Site Local addresses

Risposta

- a. Are automatically setup by IPv6 devices for on-link communications
- b. Are deprecated but they can be used in IPv6 networks
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- c. Are assigned by a central authority that guarantees their global uniqueness
- d. Do not exist

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The IPv6 address FE80::0201:06FF:FEA5:3A4C is

Risposta :

- a. An address currently not available in IPv6
- b. An address that can be used on a host with MAC address 00:01:06:A5:3A:4C for communicating with another host on the same link
- c. An address that can be used on a server with MAC address 00:01:06:A5:3A:4C to offer a service on the public IPv6 Internet
- d. An address that can be used by more than one device on the same link

RISPOSTE

- 01 C
- 02 A
- 03 A
- 04 C
- 05 D
- 06 C
- 07 C
- 08 D
- 09 B
- 10 D
- 11 C
- 12 C
- 13 C
- 14 A
- 15 A
- 16 B
- 17 C
- 18 B
- 19 B
- 20 C
- 21 A
- 22 A

23 B

24 B

25 B